Central Limit Theorem

Sampling Distribution of the Mean = E(Xbar) = μ

population each data point tells about 10,000 customers

Population				Population Mean
20	10	15	50	23.75
200000	100000	150000	500000	950000

Samples

n = 2 Combinations =4C2 6

Samples	n1	n2	Mean
S1	20	10	15
S2	20	15	17.5
S4	20	50	35
S5	10	15	12.5
S6	10	50	30
S7	15	50	32.5

Samplinng Distribution Mean 23.75 = μ

Central Limit theorem:

The sampling distribution of a sample mean becomes approximately normal as the sample size increases, provided the required conditions are satisfied.

$$SE(Xbar) = \frac{\sigma}{\sqrt{n}}$$

Conditoins for CLT

1. Random Sampling
2, Independence
3. Sufficiently large sample n> =30
4. Finite Mean and Standard Deviation

Standard Error

Standard Deviation	**Standard Error**
1. Measures variation among individual observations	Meaures variation in a sample statistic across repeated samples
2. Describes the spread of the data	Describes the precision of an estimate.
3. Often represented by σ or sd.	For the samplemean, represented by SE(Xbar)

Standard Error of the mean:

used when population standard deviation is known.

Sigma- population SD

n = sample Size

SE(Xbar) $= \frac{\sigma}{\sqrt{n}}$

Business Example

Suppose a delivery company has:

Sigma = 15 minutes

n = 100

Calculate SE

SE = sigma/SQRT(n) =15/SQRT(100)

SE =	1.5 minutes

Interpretation:

The sampling distribution of the average delivery time has a standard error of 1.5 minutes.

This doesn't mean that evry delivery is within 1.5 minutes of the average.

Standard Error of Proportion

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

Example: **Conversion Rate**

Suppose:

p =0.10, n= 400

SE = SQRT[(0.10*(1-0.10)]/400 0.015

SE = 0.015 =1.5%

Business Interpretation:

The Standard error helps analysts asses the precision of estimated conversion rates, customer sataisfaction, percentages, and defect rates.

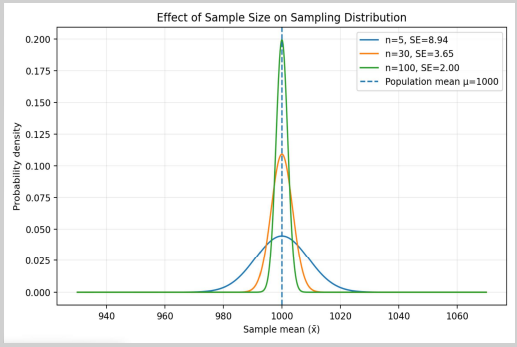
Effects of Sample Size

The Key Relationship:

$$SE = \frac{\sigma}{\sqrt{n}}$$

As sample size increases,SE decreases

Sampling Distributions



Example:

Suppose

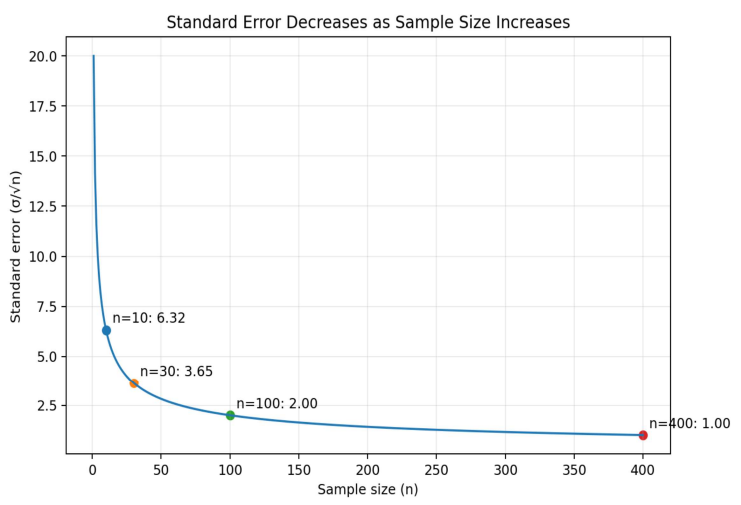
Sigma = 20

Size(n)	Calcualtion	SE
10	20/SQRT(10)	6.32
30	20/SQRT(30)	3.65
100	20/SQRT(100)	2.00
400	20/SQRT(400)	1.00

n increasing --> curve becoming narrower

Larger samples generally produce more precise sample means.

Standard Error vs Sample Size



Business Applications

1. Market Research
2. Customer Surveys
3. A/B testing
4. Muanufacturing quality control
5. Estimate average delivery time.

<u>Point Estimate vs Interval Estimate</u>

1. Point Estimate:

A single value calculated from sample data to estimate an unknown population parameter.

For the population mean: $\hat{\mu} = \bar{x}$

For a population proportion. $\hat{p} = \frac{x}{n}$

Example

A company surveys 100 customers and calcualtes average spending of ₹2,500.

What is point estimate?

Xbar = ₹2,500

It has one estimate, but it does not show the uncertaintly around that estimate.

2. Interval Estimate

A range of plausible values for an unknown population parameter, constructed using sample data and statistical method.

For a population mean when the population standard deviation is known: $\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$

For a proportion, using the usual large-sample normal approximation:

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Example: **95% confidence Interval of Mean**

Suppose:

Xbar = 30, Sigma = 10, n = 100

Standard Error: sigma/SQRT(n) 1

SE = 1

For a 95% confidence interval use z* = 1.96

CI = Xbar +- z*(SE)	
CI = 30-(1.96*1)	28.04
CI = 30+(1.96*1)	31.96
CI = (28.04,31.96)	

Interpretation

The calculated 95% confidence interval extends from 28.04 to 31.96

In repeated sampling, a correctly constructed 95% confidence interval procedure captures the true population mean in approximately 95% of sample, under its assumptions.

Confidence Intervals:

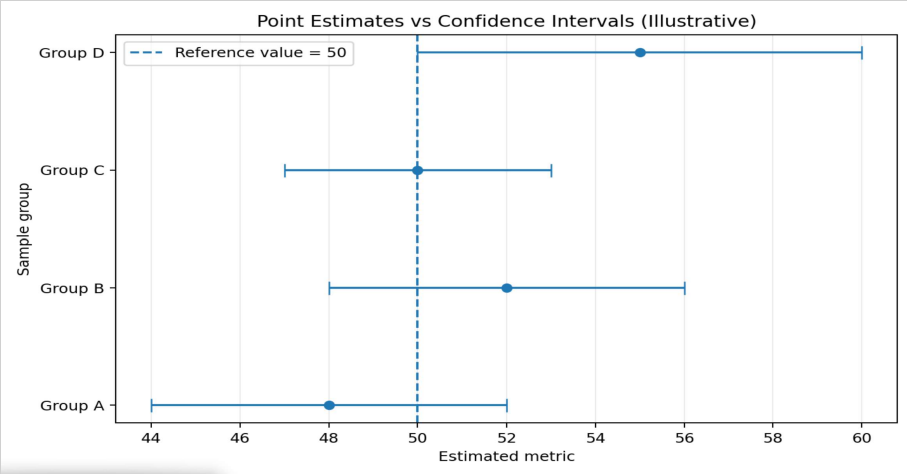
90% --> 1.645

95%--> 1.960

99%-->2.576

A higher confidence level generally produces a wider interval when the other inputs remain the same.

Point Estimate vs Confidence Interval



Question: Which gives more information: a point estimate or a confidence interval?

Answer: A confidence interval gives both a central estimate and a range that communicates uncertainty

Biased and Unbiased Estimators

1. Estimator

An estimator is a rule of statistic used to estimate an unknown population parameter.
example: **Xbar** is an estimator of the population mean μ (mu)

2. Biased Estimator

An estimator is biased when its expected value differs from the population parameter.

$$Bias(\hat{\theta}) = E(\hat{\theta}) - \theta$$

If:

$$E(\hat{\theta}) \neq \theta$$

the estimator is biased.

3. Unbiased Estimator:

an estimator is unbiased when its expected value equals the true prameter.

$$E(\hat{\theta}) = \theta$$

Bias ($\hat{\theta}$) =0

Example: -- Estimating Population Variance:

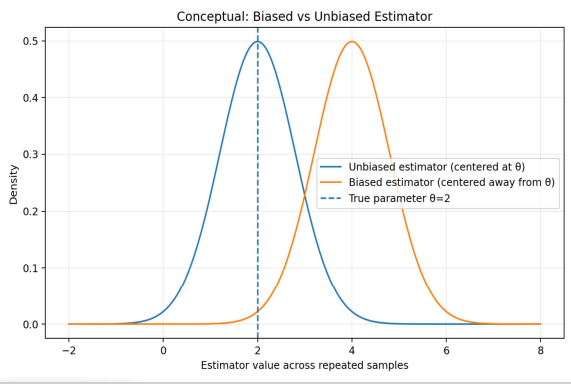
For observations x1,x2,x3,.....xn, the usual unbiased sample variance estimator is

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{X})^2}{n-1}$$

Using n instead of n-1 in denomintor gives a biased estimator of populaiton variance under ordinary random sampling from a population with finite variance.

The denominator matters when estimating population vairance from a sample.

Biased vs Unbiased estimator



Interview Point

Unbiased does not automatically mean accurate for every sample.

An unbiased estimator can still vary considerably from sample to sample. Bias and variability are different concepts.

Business Applications

1. Estimate average customer spending
2. Estimating delivery time variance
3. Market research
4. Quality control
5. A/B testing

What does a 95% confidence interval mean?

In repeated sampling, approximately 95% of intervals constructed using the same valid method would contain the true parameter.